

IEEE GAME JAM

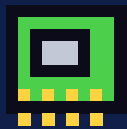
"You are the controller"

COMBO MANUAL

Pick your parts. Stick them together. Get a game. Every recipe below is one way to start. Steal it, twist it, or invent your own.

LOOT DROP: YOUR PARTS

Everything in your kit, plus what your laptop already has.



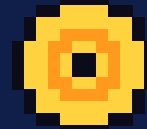
ESP32

The brain. Reads sensors, sends data to your laptop.



MPU-6050

Feels tilt, shake and swing. Great for steering and aiming.



PIEZO

Feels taps and stomps. Also beeps.



BUTTON

Fire, jump, boost, reset.



LED

Lights up to show lives, turns and hits.



CARDBOARD

Turns parts into a sword, wheel or foot pad.



LAPTOP MIC

Hear yelling. Louder = stronger.



WEBCAM

Sees your hands, face and swings.

CRAFTING RECIPES

This + this = a game you can build today. Level 1 is easiest, Level 3 is a boss fight.

RAGE QUIT SIMULATOR

LEVEL 2



Shake the controller to charge a laser. Slam the table to fire. Yell into the mic to raise a shield. The louder you are, the stronger it is.

Laptop: Python + PySerial, or a browser with the Web Audio API

DRUM HERO

LEVEL 1



Tape piezos to cardboard circles and tap to play drum sounds. Tilt the board for scratch and echo effects.

Laptop: browser + Web Audio API

STEADY HAND SAM

LEVEL 1



Guide a dot through a maze by holding the controller perfectly still. If the gyro jitters, the laptop buzzes and you lose a life.

Laptop: any language. Just compare gyro values to a threshold

TILT TANK TUSSLE

LEVEL 2



Tilt the cardboard tank to steer. Press to fire. Build two controllers for a head-to-head battle.

Laptop: Unity, Godot or Pygame with two ESP32s

CARDBOARD NINJA

LEVEL 2



Tape the sensor to a cardboard sword. Swing speed decides how hard you slice. The webcam shows you in the game.

Laptop: Python + OpenCV + PySerial

TURBO WHEEL

LEVEL 1



Build a cardboard steering wheel. Tilt to steer, one button for nitro, one for brakes.

Laptop: Unity or Godot, top-down racer

LAVA FLOOR JUMP

LEVEL 1

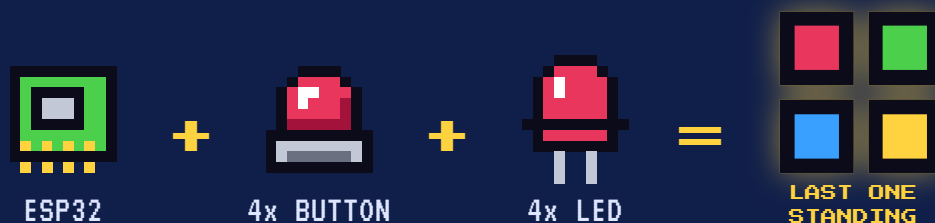


Hide a piezo under a cardboard foot pad. **Stomp** to make your character jump. A harder stomp means a higher jump.

Laptop: Godot or Pygame platformer

SIMON SAYS: BATTLE ROYALE

LEVEL 3



The laptop shows a colour sequence. Each player repeats it on their own **button box**. Miss one and your LEDs go dark.

Laptop: WebSockets or Serial, one box per player

MAKE YOUR OWN

1 INPUT (tilt / tap / press)
+ 1 OUTPUT (screen / sound / LED)
+ 1 TWIST (second sensor, mic or webcam)
= YOUR GAME

PRO TIPS

CALIBRATE

Hold the MPU-6050 still for 2 seconds at the start so the game knows where "zero" is.

DEBOUNCE

One button press can count as 10. Add a 20 ms delay or ignore repeats.

PROTECT THE PIN

Put a 1 MΩ resistor across each piezo so hard hits don't fry the ESP32.

ALWAYS HAVE RESET

Map one button to restart so a crash never ends your demo.

SOUND = 50%

A simple beep on every score makes any game feel ten times better.

START WITH USB

Get Serial working first. Go wireless only after the game is fun.

THE BRAIN: The ESP32 is in every recipe. It reads the sensors and sends numbers to your laptop over USB. The laptop runs the game.

GAME OVER? INSERT COIN TO CONTINUE

This IEEE Day · Thursday, Oct 8 · 10 AM · Rooms 12.24 and 12.25